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Flexible operation strategy of an integrated renewable multi-generation system for electricity, hydrogen, ammonia, and heating

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ABSTRACT

Sustainable development and efficient integration of different resources are imperative for future energy systems. To solve the uneven temporal–spatial distribution of renewables, green hydrogen and ammonia can be used as short- and long-term energy storage mediums, respectively. The synergistic effect of introducing these two media into an integrated renewable multi-generation system was examined in this study. The system provided a constant 10 MW electrical baseload and green ammonia when excess hydrogen was available to satisfy the electricity needs for 24 h. The effects of energy storage capacity and baseload electricity were considered. The performance of the system and strategy was thermodynamically investigated, and its generality and feasibility were dynamically evaluated based on the hourly weather data of 10 cities in China in 2020. The analysis revealed that the maximum overall energy and exergy efficiencies were 50% and 73.1%, respectively. The maximum energy efficiency in each subsystem varied from 33.7% to 73.8%, whereas the maximum exergy efficiency changed from 43.5% to 60.9%. In the baseline condition, the total energy outputs of solar, wind, and backup power subsystems were 33.3, 48.5, and 16.4 GWh, respectively. The hydrogen and ammonia yields were 994.5 and 7,201 $t \cdot y^{-1}$, respectively. The higher hydrogen and ammonia yields of Shenyang, Shanghai, Kunming, and Harbin demonstrate great potential for the development of this system.

1. Introduction

Because of the ongoing development of emerging markets and developing economies, world energy demand has increased even in the post-pandemic era. The consumption of fossil fuels has increased, and the corresponding carbon emissions have rebounded. As awareness of the harmful influence of climate change and depletion of fossil fuel reserves increases, efforts are being made to decrease the dependence on fossil fuels and increase the penetration of renewables, which is expected to account for more than half of the growth in the global electricity supply in 2021 [1]. Owing to the consistent development of renewable energy technologies, the price of solar and wind power generation has decreased; hence, renewable-energy-based infrastructures are being considered and constructed by more countries. In addition, because the power generation and industrial sectors account for most of the global carbon emissions, researchers have been investigating multi-generation systems that integrate several subsystems to provide electricity, heating, cooling, and other products. They strive to construct a 100% renewable system that increases the overall energy efficiency and solves the high energy consumption and carbon emissions in some industries, such as ammonia synthesis and iron manufacturing. Analytical and mathematical models have been designed and analyzed in terms of technical, financial, and environmental factors [2]. To cope with the seasonal and intermittent features of renewable energy and flexibility requirements, energy storage is an important technology introduced in many multi-generation systems. An energy storage system converts excess electricity into other forms of energy and then converts it back to electricity when required. Considering the technical characteristics, costs, and environmental effects, thermal and hydrogen storage are suitable for bulk energy storage [3]. Thermal storage units in solar thermal power plants have a relatively low capital cost, mature technology, and high efficiency and stabilize fluctuating solar energy and can provide district heating. Hydrogen storage has been considered a promising energy carrier in recent years because hydrogen, regarded as a substitute for fossil fuels, can be used as both fuel and feedstock in many different applications. In addition, the gravimetric energy density of hydrogen is higher than that of conventional fossil fuels. However, the low volumetric density and high flammability of hydrogen make it challenging to adopt for energy storage and as a transportation fuel.

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Nomenclature

A	area, m^2
C	axial induction factor, -
d	diameter, mm
E	voltage, V
$\dot{E}x$	exergy destruction rate, $kJ \cdot kg^{-1}$
ex	specific exergy, $kJ \cdot kg^{-1}$
F	Faraday constant, $C \cdot mol^{-1}$
F_R	collector removal factor, -
h	specific enthalpy, $kJ \cdot kg^{-1}$
h_w	heat transfer coefficient, $W \cdot m^{-2} \cdot K^{-1}$
$\dot{I}$	solar irradiance, $W \cdot m^{-2}$
J	current density, $A \cdot m^{-2}$
L	thickness of membrane, mm
M	mole mass, $g \cdot mol^{-1}$
$\dot{m}$	mass flow rate, $kg \cdot s^{-1}$
N	number, -
$\dot{N}$	mole rate, $mol \cdot s^{-1}$
P	pressure, MPa
$\dot{Q}$	heat transfer rate, kW
R	ideal gas constant, -
R_g	gas constant, -
$\dot{S}$	entropy generation rate, $kJ \cdot kg^{-1} \cdot K^{-1}$
s	specific entropy, $kJ \cdot kg^{-1} \cdot K^{-1}$
T	temperature, ℃
U_L	heat loss coefficient, $W \cdot m^{-2} \cdot K^{-1}$
V	wind speed, $m \cdot s^{-1}$
$\dot{W}$	power, kW
W	width, m
z	height, m

Greek letters

α	wind shear exponent, -
ε	emissivity, -
η	efficiency, -
λ	membrane water content, -
$\overline{\rho}$	average air density,$kg \cdot m^{-3}$
σ	Stefan–Boltzmann constant, $W \cdot m^{-2} \cdot K^{-4}$

Subscripts

a	ambient
an	anode
act	activation
ap	aperture
b	beam
c	compressor
ca	cathode
cell	fuel cell
con	concentration
d	diffuse
dest	destruction
en	energy
ex	exergy
gen	generation
hx	heat exchanger
m	mechanical
me	meteorological
in	inlet
iso	isothermal
opt	peak optical
out	outlet
p	pump
r	receiver
ro	rotor
t	turbine
ts	thermal storage
w	wind output
WT	wind turbine

Acronyms

HTF	heat transfer fluid
LHV	lower heating value
PEM	proton exchange membrane
PSA	pressure swing adsorption

Existing gas-, liquid-, and solid-state hydrogen storage methods still face several technical challenges such as high energy loss and immature hydrogen storage materials [4].

To efficiently and safely store hydrogen, ammonia has recently been proposed as a hydrogen storage medium because the ammonia molecule contains three hydrogen atoms and has a higher hydrogen content compared to that of other storage media. In addition, it has a high volumetric density and more stable physical characteristics and can be stored in liquid form at ambient temperatures and moderate pressure [5]. Moreover, it is practically well known that ammonia is a zero-carbon fuel that can be used in internal combustion engines, boilers, and fuel cells, making it a promising energy storage medium [6]. Traditional ammonia synthesis is performed using fossil fuels, mostly natural gas and coal, which are energy-intensive and have a high carbon emission level. In contrast, green ammonia production, in which the feedstock comes from renewable-driven water electrolysis and air separation, is seen as a clean route for ammonia production. The ammonia market is of great interest in small-scale green ammonia plants [6]. If renewable energy-based power systems are equipped with small-scale green ammonia plants, they can satisfy both the requirements of clean production of ammonia and the need for energy storage. Therefore, it is imperative to evaluate this win–win option.

Some research has emphasized the technical and economic analyses of green ammonia production, where green ammonia mostly served as a product. Sánchez et al. proposed a renewable energy-driven ammonia plant with two cooling designs. They focused on the detailed design of the production process and planning; renewable energy was less involved and was only considered in the economic analysis and operation optimization [7]. Allman et al. developed a scheduling model for green ammonia synthesis. By dynamically controlling each subsystem's capacity and production rate based on weather forecasts and energy prices, the minimum operation cost was found in the case study, and factors affecting the operation cost and its relationships were also found [8]. Moreover, some researchers have considered green ammonia as an energy storage medium in energy systems. Rouwenhorst et al. conceptualized a power-to-ammonia-to-power process in an energy system including wind turbines, solar panels, and fuel cells. The round-trip efficiency was 61%, and the operational cost was 0.3–0.35 €·kWh^{-1} [9]. Ikäheimo et al. evaluated the power-to-ammonia strategy in a simulation model based on North European power and district heat systems. They performed both dispatch scheduling and optimization of capital cost. They found that ammonia as a long-term energy storage medium had less impact on the system cost, and the production cost of ammonia did not significantly exceed the world market price, proving that power-to-ammonia is a cost-effective strategy [10]. Palys et al. proposed an ammonia-based sustainable energy agriculture system that used renewable energy (solar and wind power) to produce ammonia used in different agricultural applications. The system was designed to meet the

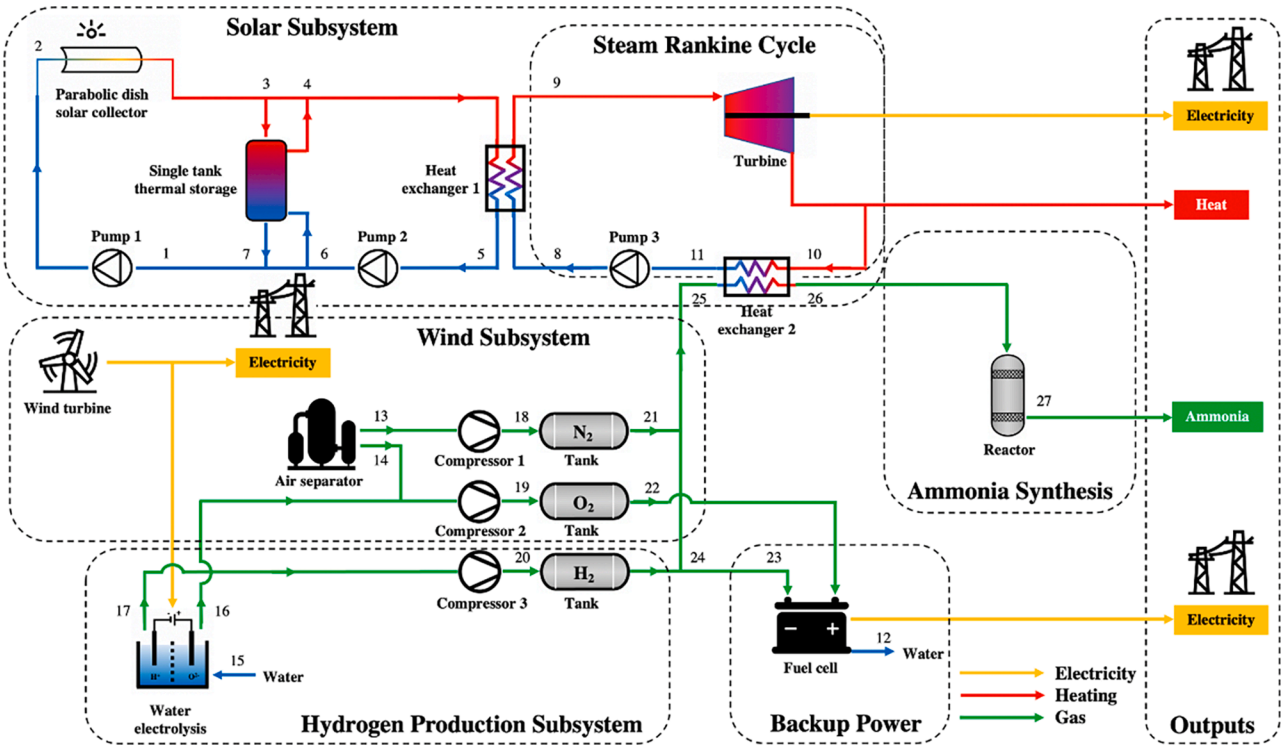


Fig. 1. Schematic of integrated renewable multi-generation system.

local electricity demand and to export predictable and consistent power to a grid. It was constructed based on the synergies between ammonia production and energy supply, and its economic viability was evaluated using case studies [11].

Furthermore, thermodynamic models of multi-generation systems that involved green ammonia production have been constructed to evaluate the performance of an integrated system and all subsystems in detail. Hasan et al. proposed a bi-generation system with wind turbines, solar panels, water electrolyzers, and ammonia synthesis reactors based on a steady-state model. The total energy and exergy efficiencies at maximum power output were 18.8% and 19.1%, respectively. Through a sensitivity analysis, the effects of wind speed and solar irradiance on efficiencies, production rates of hydrogen and ammonia, and exergy destruction were evaluated [12]. Shahid et al. constructed an independent multi-generation system for electricity, heating, cooling, drinking water, and ammonia. The total energy and exergy efficiencies were 18.9% and 28.0%, respectively. Sensitivity analysis was conducted based on various solar irradiances, ambient temperatures, and inlet pressures. In addition, it was found that the multi-generation energy system reduced carbon emissions by 60% compared to a non-renewable system [13]. Siddiqui et al. designed a solar and wind energy-based system for electricity, heating, hydrogen, and ammonia. In the system, excess wind energy was used to produce hydrogen and ammonia, while the heat from solar energy was used for power generation and ammonia dissociation. A dynamic simulation was performed to evaluate the average system performance on the days of each month. The total energy efficiency varied between 37.4% and 62.9%, while the total exergy efficiency varied between 35.2% and 55.5% [14]. They further focused on the performance of integrated solar-based ammonia synthesis and fuel cell systems [15] and a solar and wind energy-based system equipped with a reverse osmosis desalination subsystem [16].

Previous studies have discussed green ammonia production in terms of the technical and economic aspects and considered it as the energy storage method in an energy system. However, few studies have evaluated systems based on dynamic simulation; hence, the energy and mass flow in the systems were not well constructed. They omitted the capacity of storage tanks and the need for a constant electricity output. In addition, the operation strategy should be more flexible and consider both output and storage capacity. In this study, a flexible operation strategy and an integrated renewable multi-generation energy system were designed using a thermodynamic method. The performance was evaluated based on the hourly meteorological data of 10 cities in China. The system used hydrogen as a short-term energy storage method to mitigate the uneven distribution of renewable energy within one day and ammonia as a long-term energy storage method to solve the seasonal differences in renewable energy. It consisted of a solar subsystem equipped with a 9-h thermal storage tank and steam turbine, a wind subsystem, a hydrogen production subsystem equipped with a water electrolyzer and storage tanks, a backup power subsystem equipped with fuel cells, and an ammonia synthesis subsystem equipped with an air separation unit. This strategy considered the effects of the capacity of thermal storage and hydrogen storage.

The innovative objectives of this study were: 1) to design an integrated renewable multi-generation energy system and relevant flexible operation strategy to provide a constant electricity baseload (10 MW), district heating, hydrogen, and green ammonia; 2) to evaluate the performance of the system and its subsystems based on dynamic simulation through the energy and exergy efficiencies and the yields of hydrogen and green ammonia; and 3) to compare the potential of the system in the selected cities and make further suggestions.

2. System description and analyses

This section introduces the detailed information of the system design, flexible operation strategy, analytical model of each subsystem, subsystem efficiencies, and meteorological data. The performance of the overall system and its components was evaluated using the energy and exergy analyses tool. The steady-state model was constructed and performed using the Engineering Equation Solver (F-Chart software, LLC) [17]. Then, the flexible operation strategy was realized by a self-made code, and a dynamic simulation was conducted on the hourly meteorological data of 10 cities in China in 2020. During the modeling, the

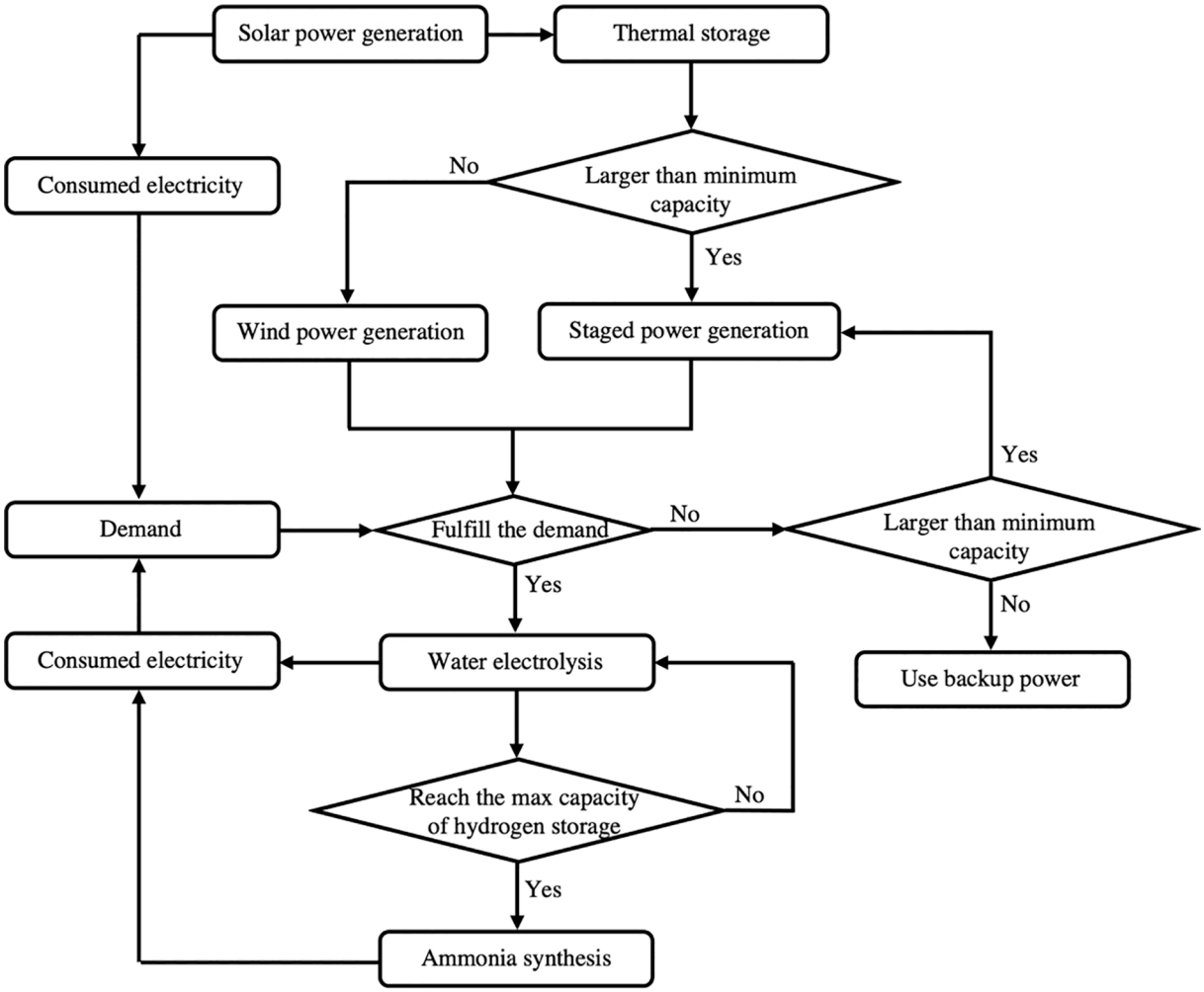


Fig. 2. Flow chart of flexible operation strategy.

following assumptions were made: 1) the ambient temperature and pressure were 25 °C and 101.3 kPa, respectively; 2) because the flow speed was relatively low and there was no obvious height difference, the potential and kinetic energy and exergy were neglected; 3) the pumps and steam turbine operated adiabatically, while the compressors operated isothermally; 4) the mass, pressure, and heat losses in the pipelines were negligible; 5) because the time resolution was 1 h in the meteorological data, the solar irradiance and wind speed were constant in each time interval; and 6) for easy analysis, each subsystem simplified the purification process by omitting ammonia synthesis.

2.1. Integrated renewable multi-generation system

The integrated renewable multi-generation system is shown in Fig. 1. There were three loops in the solar subsystem. At State 1, the heat transfer fluid (HTF) enters Pump 1 to increase its pressure to the rated pressure. Then, it is sent to the parabolic trough collector at State 2. After absorbing solar irradiation, it reaches the designated temperature at State 3 and enters the thermal storage tank used to stabilize the intermittent solar energy and store excess energy. The cold HTF returns to Pump 1 at State 7. Pump 2 drives another loop to provide a stable heat output. The hot HTF enters Heat Exchanger 1 at State 4 to heat up the superheated steam that entered the steam turbine at State 9 to generate electricity. The cooled HTF is sent back to the thermal storage tank at State 6. In the Rankine cycle, Pump 3 provides the driving force for water circulation. The waste heat at State 10 is used for district heating or to preheat the hydrogen and nitrogen mixture in Heat Exchanger 2 before being sent to the ammonia synthesis reactor.

The excess electricity from the wind turbines is used for proton exchange membrane (PEM) water electrolysis and pressure swing adsorption (PSA) air separation in the wind subsystem. Air enters the PSA unit to generate nitrogen and oxygen at States 13 and 14, respectively. In the hydrogen production subsystem, water enters the PEM electrolyzer at State 15 to produce hydrogen and oxygen at States 16 and 17, respectively. Nitrogen, oxygen, and hydrogen are compressed into storage tanks in States 18–20. The compressors work isothermally, and the released heat is used for district heating. For simplification, the detailed heat flow is not presented here. The backup power subsystem is activated when there is an energy shortage. Hydrogen at State 23 and oxygen at State 22 are used in the PEM fuel cell. The water in State 12 can be reused in water electrolysis. In the ammonia synthesis subsystem, nitrogen at State 21 and hydrogen at State 24 are mixed at a mole ratio of 1:3 and sent to Heat Exchanger 2 at State 25. After preheating, the mixture enters the synthesis reactor at State 26, and the liquid ammonia is stored for later use at State 27.

2.2. Flexible operation strategy

The flexible operation strategy was designed to provide a constant baseload electricity, district heating, hydrogen, and green ammonia. Thermal and hydrogen storage systems play an important role as short-term energy storage methods (within one day), while green ammonia as a long-term energy storage method mitigates the seasonal differences of renewable energy and is a product to be exported. A flowchart of the

Table 1
Parameters of solar subsystem [23].

Name	Symbol	Value	Unit
Parabolic trough collector (type)		Ultimate trough	
Receiver diameter	d_r	70	mm
Aperture width	W_{ap}	7.5	m
Aperture area	A_{ap}	150,000	m^2
Inlet temperature	T_{in}	290	℃
Outlet temperature	T_{out}	565	℃
Working pressure in receiver	P_r	1,200	kPa
Peak optical efficiency	η_{opt}	0.75	
Collector removal factor	F_R	0.9	
Emissivity of receiver	ε_r	0.86	
HTF (type)		Solar salt	
Max thermal storage capacity		9	h
Pump isentropic efficiency	η_p	0.9	
Steam turbine isentropic efficiency	η_t	0.8	
Inlet pressure of Rankine cycle		13,000	kPa

strategy is shown in Fig. 2.

The strategy starts with solar power generation. In the solar subsystem, the hot HTF from the receiver first enters the thermal storage tank. Then, if the thermal storage capacity denoted by the total weight of the hot HTF is larger than the minimum capacity (1-h operation), the staged output power from the Rankine cycle is determined according to the capacity. If the capacity is less than the minimum, the output is zero, and the wind power generation is the final output. If the summation of wind and staged power generation can fulfill the demand, excess electricity is used for water electrolysis of the hydrogen production subsystem.On the other hand, if it cannot satisfy the demand, the remaining thermal storage capacity will be reevaluated. If it is larger than the minimum, which means that it still has the potential to provide more energy, the staged output power will be increased. Subsequently, the total output power is estimated again until it can fulfill the demand or the capacity is less than the minimum. At this time, the backup power subsystem is activated to offset the energy shortage. In the hydrogen production subsystem, if the hydrogen storage reaches the maximum capacity (24-h operation), the excess hydrogen will be exported to the ammonia synthesis subsystem. After accumulating sufficient hydrogen, ammonia is produced at a constant rate, and the required nitrogen is simultaneously produced in the PSA unit. In addition, the electricity consumed by pumps, compressors, and the PSA unit is added to the total demand.

Production priority is important for this strategy. The constant baseload electricity is the primary output fulfilled by the solar, wind, and backup power subsystems. Thermal storage ensures that the solar subsystem provides stable output power as the baseload of the total output power, particularly in the daytime. Wind energy fluctuates more; hence, hydrogen plays an important role in storing excess wind energy and offsetting the energy shortage in a short time. Green ammonia is a subordinate output. For safety and cost considerations, the storage capacity of hydrogen is limited. When energy is abundant, the produced hydrogen exceeds the maximum storage capacity, and a portion of hydrogen is exported to the ammonia synthesis subsystem. Considering the efficiency and lifetime, the ammonia synthesis subsystem cannot work intermittently but only periodically. When the amount of accumulated hydrogen satisfies a constant production rate over a period of time, the subsystem starts to operate. District heating is the final output. The waste heat from the steam turbine and the heat released from the compressors, PSA unit, water electrolysis, fuel cell, and ammonia synthesis reactor are collected for district heating. The heat demand is not restricted here.

Table 2
Parameters of wind subsystem [25–27].

Name	Symbol	Value	Unit
Axial induction factor	C	0.49	
Efficiency of a single wind turbine	η_{WT}	0.9	
Number of wind turbines	N_{WT}	12	
Average air density	$\overline{\rho}$	1.255	$kg{\cdot}m^{-3}$
Rotor diameter	d_{ro}	82	m
Height of rotor	z_{ro}	71.5	m
Wind shear exponent	α	0.25	
Mechanical efficiency	η_m	0.9	
Isothermal efficiency	η_{iso}	0.9	
Inlet pressure of PSA	P_{in}	40	kPa
Outlet pressure of PSA	P_{out}	200	kPa
Pressure of storage tanks		15	MPa

2.3. Solar subsystem

The analytical models of the parabolic trough collector, thermal storage tank, heat exchanger, pumps, and Rankine cycle were constructed based on references [18–22]. The detailed information is shown in Table 1. The thermodynamic analysis is presented in the **Appendix**.

Through Eqs. (1–4), the effective received energy and energy losses in the parabolic trough collector are determined. The total solar irradiance is expressed as

$$\dot{I} = (\dot{I}_b + \dot{I}_d)\eta_{opt} \tag{1}$$

where $\dot{I}$ is the total solar irradiance reflected by the collector [$W{\cdot}m^{-2}$], $\dot{I}_b$ is the incoming beam irradiance [$W{\cdot}m^{-2}$], $\dot{I}_d$ is the incoming diffuse irradiance [$W{\cdot}m^{-2}$], and η_{opt} is the peak optical efficiency of the collector [-].

The effective energy obtained by the receiver is determined as

$$\dot{Q}_{solar} = F_R A_{ap}\left[\dot{I} - \frac{A_r}{A_{ap}}U_L(T_r - T_a)\right] \tag{2}$$

where $\dot{Q}_{solar}$ is the effective energy [W], F_R is the collector removal factor that considers the effect of heat conduction in the receiver [-], A_{ap} is the aperture area [m^2], A_r is the receiver area [m^2], U_L is the heat loss coefficient [$W{\cdot}m^{-2}{\cdot}K^{-1}$], T_r is the receiver temperature [K], and T_a is the ambient temperature [K].

The receiver temperature was the average of the inlet and outlet temperatures since the conduction between the HTF and receiver was not considered. The heat loss of the receiver had two components: convection and radiation. The heat loss coefficient was estimated according to the following equations:

$$\dot{Q}_{loss} = A_r h_w(T_r - T_a) + A_r\varepsilon_r\sigma(T_r^4 - T_a^4) \tag{3}$$

$$\dot{Q}_{loss} = U_L A_r(T_r - T_a) \tag{4}$$

where $\dot{Q}_{loss}$ is the heat loss of the receiver [W], h_w is the convective heat transfer coefficient [$W{\cdot}m^{-2}{\cdot}K^{-1}$], ε_r is the emissivity of the receiver, [-], and σ is the Stefan–Boltzmann constant.

The energy and exergy efficiencies of the solar subsystem are

$$\eta_{en,\ solar} = \frac{\int \dot{W}_t dt}{\int \dot{m}_4 h_4 dt - \int \dot{m}_5 h_5 dt} \tag{5}$$

$$\eta_{ex,\ solar} = \frac{\int \dot{W}_t dt}{\int \dot{m}_4 ex_4 dt - \int \dot{m}_5 ex_5 dt} \tag{6}$$

2.4. Wind subsystem

The analytical model of the wind subsystem including wind turbines,

Table 3
Parameters of hydrogen production subsystem.

Name	Symbol	Value	Unit
Faraday constant	F	96,486	$C{\cdot}mol^{-1}$
Operating temperature of PEM	T_{PEM}	80	℃
Thickness of membrane	L_{PEM}	100	μm
Current density	J_{PEM}	6,000	$A{\cdot}m^{-2}$
Exchange current density of anode	$J_{0,\,an}$	10^{-5}	$A{\cdot}m^{-2}$
Exchange current density of cathode	$J_{0,\,ca}$	10	$A{\cdot}m^{-2}$
Charger transfer coefficient of anode	α_{an}	0.5	
Charger transfer coefficient of cathode	α_{ca}	0.5	
Lower heating value (LHV) of hydrogen	LHV_{H_2}	1.21×10^5	$kJ{\cdot}kg^{-1}$
Anode interface	λ_a	14	μm
Cathode interface	λ_c	10	μm

air separators, compressors, and gas storage tanks was constructed according to references [24–28]. The detailed information is shown in Table 2, and the thermodynamic analysis is presented in the **Appendix**.

The output power of wind turbines can be estimated using the kinetic energy of the wind as

$$\dot{W}_{wind} = \frac{N_{WT}}{2} C\eta_{WT}\overline{\rho}A_{WT}V^3 \tag{7}$$

where $\dot{W}_{wind}$ is the effective output power of the wind turbines [W], N_{WT} is the number of wind turbines [-], C is the axial induction factor denoting the power generation efficiency [-], η_{WT} is the efficiency of a single wind turbine [-], $\overline{\rho}$ is the average air density [$kg{\cdot}m^{-3}$], A_{WT} is the rotor area of the wind turbine [m^2], and V is the wind speed [$m{\cdot}s^{-1}$]. The effects of wind direction, temperature, and humidity were not considered in this study.

The actual wind speed at the height of the rotor needs needs to be corrected as

$$V = V_{me}\left(\frac{z_{ro}}{z_{me}}\right)^{\alpha} \tag{8}$$

where V_{me} is the wind speed measured at the meteorological station [$m{\cdot}s^{-1}$], z_{ro} is the height of the rotor [m], z_{me} is the height of the measuring instrument [m], and α is the wind shear exponent [-].

PSA air separation technology was used to produce the oxygen and nitrogen used in the fuel cells and ammonia synthesis, respectively. Air was first compressed at high pressure and then sent to two reactors equipped with zeolites and carbon molecular sieves. The input power was determined by an isothermal process as

$$\dot{W}_{PSA} = \frac{\dot{m}_{air}}{\eta_m\eta_{iso,\,PSA}} T_{PSA}R_g\ln\frac{P_{out}}{P_{in}} \tag{9}$$

where $\dot{W}_{PSA}$ is the input power of the air separator [W], $\dot{m}_{air}$ is the mass flow rate of air [$kg{\cdot}s^{-1}$], η_m is the mechanical efficiency [-], $\eta_{iso,\,PSA}$ is the isothermal efficiency [-], T_{PSA} is the working temperature of air separation [K], and P is the outlet and inlet pressure [kPa].

The input power of the compressors was calculated as

$$\dot{W}_c = \frac{\dot{m}_c}{\eta_m\eta_{iso,\,c}} T_cR_g\ln\frac{P_{out}}{P_{in}} \tag{10}$$

where $\dot{W}_c$ represents the input power of the compressors [W], $\dot{m}_c$ is the mass flow rate [$kg{\cdot}s^{-1}$], and T_c is the working temperature of the compressors [K].

The energy and exergy efficiencies of the wind subsystem are

$$\eta_{en,\,wind} = \eta_{ex,\,wind} = \frac{\int \dot{W}_w dt}{\int \dot{W}_{wind} dt} \tag{11}$$

where $\dot{W}_w$ is the output power of the wind subsystem [W].

2.5. Hydrogen production subsystem

The hydrogen production subsystem was activated when excess electricity was available. PEM water electrolysis was used to generate hydrogen. The analytical model was constructed based on previous studies [29,30]. The detailed information is shown in Table 3, and the thermodynamic analysis is presented in the **Appendix**.

The electricity consumed by PEM water electrolysis was determined as

$$\dot{W}_{PEM} = J_{PEM}E_{PEM}A_{PEM} \tag{12}$$

where $\dot{W}_{PEM}$ is the input power [W], J_{PEM} is the current density [$A{\cdot}m^{-2}$], E_{PEM} is the working voltage of electrolysis (V), and A_{PEM} is the membrane area (m^2).

The working voltage was calculated as

$$E_{PEM} = E_0 + E_{act} + E_{ohm} + E_{con} \tag{13}$$

where E_0 is the reversible overpotential determined by the Nernst equation [V], E_{act} is the activation overpotential of the anode and cathode [V], E_{ohm} is the ohmic overpotential of the electrolyte [V], and E_{con} is the concentration overpotential omitted here because of the relatively low current density.

The reversible overpotential can be written as

$$E_0 = E^0 + \frac{RT_{PEM}}{2F}\left(\frac{p_{H_2}\cdot p_{O_2}^{0.5}}{P_{H_2O}}\right) = 1.23 - \frac{0.85}{1000}(T_{PEM} - 298) \tag{14}$$

where E^0 is the cell potential [V], T_{PEM} is the working temperature [K], F is the Faraday constant [$C{\cdot}mol^{-1}$], and P_{H_2} and P_{O_2} are the partial pressures [kPa].

The activation overpotential was used to activate the electrochemical reactions at the anode and cathode expressed as the Butler–Volmer equation,

$$E_{act} = \frac{RT_{PEM}}{2\alpha_{an}F}ln\left(\frac{J_{PEM}}{J_{0,\,an}}\right) + \frac{RT_{PEM}}{2\alpha_{ca}F}ln\left(\frac{J_{PEM}}{J_{0,\,ca}}\right) \tag{15}$$

where $J_{0,i}$ is the exchange current density [$A{\cdot}m^{-2}$], α is the charge transfer coefficient of the anode and cathode, and subscripts 'an' and 'ca' represent the anode and cathode, respectively.

The ohmic overpotential was determined by the resistance and current density as

$$E_{ohm} = J_{PEM}R_{PEM} \tag{16}$$

where R_{PEM} is the resistance of electrolytes determined by the local ionic conductivity [Ω].

The empirical equation of local ionic conductivity is

$$\sigma(\lambda(x)) = (0.5139\lambda(x) - 0.326)exp\left(1268\left(\frac{1}{303} - \frac{1}{T_{PEM}}\right)\right) \tag{17}$$

$$\lambda(x) = \frac{\lambda_{an} - \lambda_{ca}}{L}x + \lambda_{ca} \tag{18}$$

where $\sigma(\lambda(x))$ is the local ionic conductivity, $\lambda(x)$ is the membrane water content at location ×, and λ_{an} and λ_{ca} are the anode and cathode interfaces, respectively [μm].

The total resistance is expressed as

$$R_{PEM} = \int_0^{L_{PEM}} \frac{1}{\sigma(\lambda(x))}dx \tag{19}$$

The mole rate of hydrogen production can be written as

$$\dot{N}_{H_2} = \frac{J_{PEM}}{2F} \tag{20}$$

$$\dot{m}_{17} = \dot{N}_{H_2}M_{H_2}A_{PEM} \tag{21}$$

$$\dot{m}_{16} = 0.5\dot{N}_{H_2}M_{O_2}A_{PEM} \tag{22}$$

Table 4
Parameters of backup power subsystem.

Name	Symbol	Value	Unit
Operating temperature	T_{cell}	80	℃
Thickness of membrane	L_{cell}	178	μm
Current density	J_{cell}	15,000	$A{\cdot}m^{-2}$
Limiting current density	J_L	20,000	$A{\cdot}m^{-2}$
Exchange current density of anode	$J_{0,\,an}$	0.089	$A{\cdot}m^{-2}$
Exchange current density of cathode	$J_{0,\,ca}$	1,420	$A{\cdot}m^{-2}$
Hydrogen pressure		100	kPa
Oxygen pressure		100	kPa

where $\dot{N}_{H_2}$ is the molar rate of hydrogen production [$mol{\cdot}s^{-1}$], M_{H_2} is the mole mass of hydrogen [$g{\cdot}mol^{-1}$], and M_{O_2} is the mole mass of oxygen [$g{\cdot}mol^{-1}$].

The energy and exergy efficiencies of the hydrogen production subsystem are

$$\eta_{en,\,hydrogen} = \frac{\int \dot{m}_{17} LHV_{H_2} dt}{\int \dot{W}_{PEM} dt} \tag{23}$$

$$\eta_{ex,\,hydrogen} = \frac{\int \dot{m}_{17} ex_{17} dt}{\int \dot{W}_{PEM} dt} \tag{24}$$

2.6. Backup power subsystem

The backup power subsystem is activated when there is an energy shortage. The PEM fuel cell model was constructed based on references [31–33]. The detailed information is shown in Table 4, and the thermodynamic analysis is presented in the **Appendix**.

The output power is determined by

$$\dot{W}_{cell} = N_{cell} E_{cell} J_{cell} A_{cell} \tag{25}$$

where $\dot{W}_{cell}$ is the output power [W], N_{cell} is the number of stacks [-], E_{cell} is the working voltage of the fuel cell [V], J_{cell} is the current density [$A{\cdot}m^{-2}$], and A_{cell} is the membrane area [m^2].

The working voltage is denoted as

$$E_{cell} = E_0 - E_{act} - E_{ohm} - E_{con} \tag{26}$$

where E_0 is the reversible cell potential [V], E_{act} is the activation polarization (reaction rate loss) [V], E_{ohm} is the ohmic polarization (resistance rate loss) [V], and E_{con} is the concentration polarization (gas transport loss) [V].

According to the Nernst equation, the reversible cell potential can be expressed by the empirical equation

$$E_0 = 1.23 - 0.85 \times 10^{-3}(T_{cell} - 298.15) + 4.31 \times 10^{-5} ln\left(P_{H_2}\sqrt{P_{O_2}}\right) \tag{27}$$

where T_{cell} is the working temperature [K].

The activation polarization can be determined by

$$E_{act} = \frac{RT_{cell}}{2\alpha_{an}F} ln\left(\frac{J_{cell}}{J_{0,\,an}}\right) + \frac{RT_{cell}}{2\alpha_{ca}F} ln\left(\frac{J_{cell}}{J_{0,\,ca}}\right) \tag{28}$$

The Ohmic polarization can be determined by

$$E_{ohm} = J_{cell} R_{cell} \tag{29}$$

where R_{cell} is the total resistance determined using Eqs. (59–61)

The concentration polarization can be determined by

$$E_{con} = \frac{RT_{cell}}{2F} ln\left(1 - \frac{J_{cell}}{J_L}\right) \tag{30}$$

where J_L is the limiting current density [$A{\cdot}m^{-2}$].

The mole rate of hydrogen consumption can be written as

$$\dot{N}_{H_2} = \frac{J_{cell}}{2F} \tag{31}$$

$$\dot{m}_{23} = \dot{N}_{H_2} M_{H_2} A_{cell} \tag{32}$$

$$\dot{m}_{22} = 0.5 \dot{N}_{H_2} M_{O_2} A_{cell} \tag{33}$$

The energy and exergy efficiencies of the backup power subsystem are

$$\eta_{en,\,cell} = \frac{\int \dot{W}_{cell} dt}{\int \dot{m}_{23} LHV_{H_2} dt} \tag{34}$$

$$\eta_{ex,\,cell} = \frac{\int \dot{W}_{cell} dt}{\int \dot{m}_{23} ex_{23} dt} \tag{35}$$

2.7. Ammonia synthesis subsystem

The ammonia synthesis subsystem is activated when the amount of accumulated hydrogen is sufficient to produce ammonia over a period of time. The chemical reaction is reversible and exothermic; therefore, a suitable temperature and pressure for ammonia synthesis are necessary [34]. The working temperature was 500 °C and the working pressure was 15 MPa. The thermodynamic analysis is presented in the **Appendix**.

The energy and exergy efficiencies of the ammonia synthesis subsystem are

$$\eta_{en,\,ammonia} = \frac{\int \dot{m}_{27} LHV_{NH_3} dt}{\int \dot{Q}_{hx2} dt + \int \dot{m}_{26} h_{26} dt} \tag{36}$$

$$\eta_{ex,\,ammonia} = \frac{\int \dot{m}_{27} ex_{27} dt}{\int \dot{Q}_{hx2}\left(1 - \frac{T_a}{T_{26}}\right) dt + \int \dot{m}_{26} ex_{26} dt} \tag{37}$$

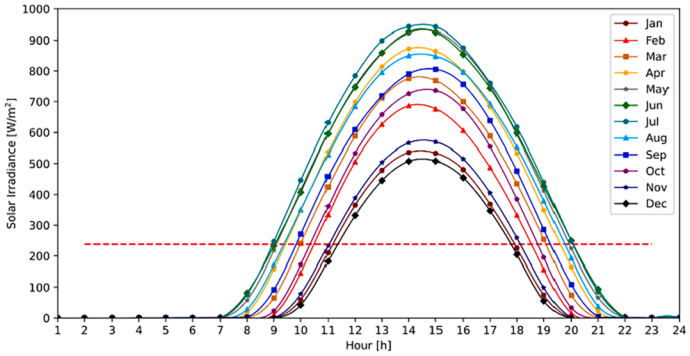
2.8. Overall efficiencies

The overall energy and exergy efficiencies of the entire system are

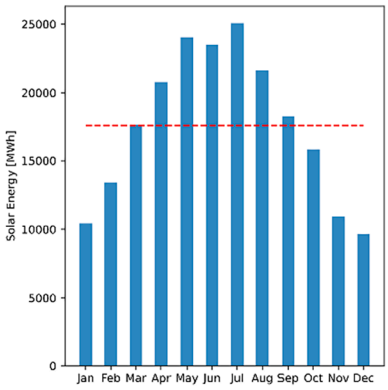
Table 5
Meteorological data of cities.

Cities	Longitude	Latitude	Max wind speed [$m{\cdot}s^{-1}$]	Average wind speed [$m{\cdot}s^{-1}$]	Max solar irradiance [$W{\cdot}m^{-2}$]	Average solar irradiance [$W{\cdot}m^{-2}$]
Beijing	39.48° N	116.28° E	21.9	5.22	1084.5	235.65
Guangzhou	23.13° N	113.29° E	15.44	5.38	1168	264.3
Harbin	45.56° N	126.34° E	23.37	6.35	1065.5	197.35
Kunming	25.00° N	102.39° E	16.59	5.89	1163.5	279.86
Shanghai	31.24° N	121.27° E	21.66	6.32	1138.5	243.49
Shenyang	41.44° N	123.31° E	17.5	6.36	1082.5	223.55
Urumqi	43.47° N	87.39° E	16.78	4.97	1065.5	214.25
Wenjiang	30.45° N	103.52° E	15.92	3.78	1146.5	245.95
Wuhan	30.36° N	114.03° E	17.34	4.9	1143.5	241.72
Yuzhong*	**35.52° N**	**104.09° E**	**15.74**	**4.95**	**1088**	**237.88**

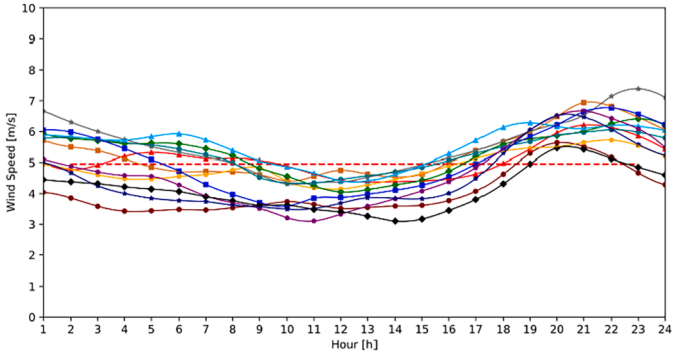
* Yuzhong was chosen for the demonstration.



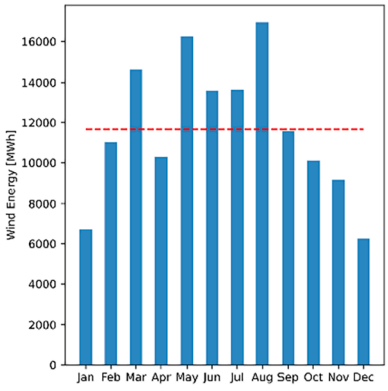
(a) Daily distribution of solar irradiance evaluated on the monthly average of each hour



(b) Total monthly solar energy



(c) Daily distribution of wind speed evaluated on the monthly average of each hour



(d) Total monthly wind energy

Fig. 3. Daily average and total monthly distribution of solar and wind energy.

$$\eta_{en} = \frac{\int \dot{W}_t dt + \int \dot{W}_w dt + \int \dot{W}_{cell} dt - \int \dot{W}_{consumed} dt + \int \dot{m}_{27} LHV_{NH_3} dt}{\left(\int \dot{m}_4 h_4 dt - \int \dot{m}_5 h_5 dt\right) + \int \dot{W}_{wind} dt} \quad (38)$$

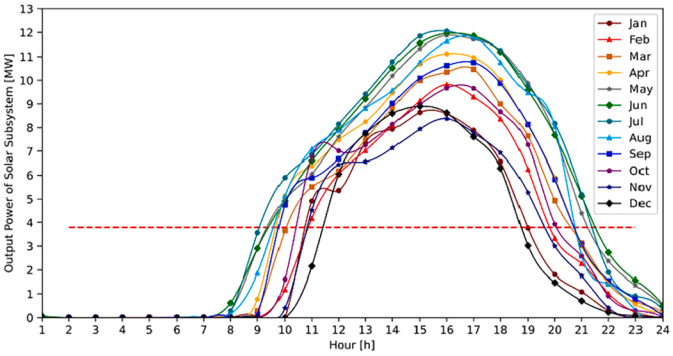
$$\eta_{ex} = \frac{\int \dot{W}_t dt + \int \dot{W}_w dt + \int \dot{W}_{cell} dt - \int \dot{W}_{consumed} dt + \int \dot{m}_{27} ex_{27} dt}{\left(\int \dot{m}_4 ex_4 dt - \int \dot{m}_5 ex_5 dt\right) + \int \dot{W}_{wind} dt} \quad (39)$$

where $\dot{W}_{consumed}$ is the consumed electricity of the entire system [W].

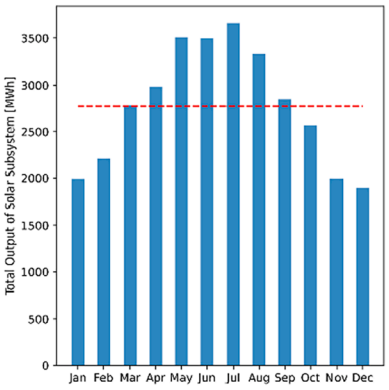
2.9. Meteorological data

The net solar irradiance, wind speed, and ambient temperature in 2020 were obtained from the MERRA-2 database provided by the Global Modeling and Assimilation Office, Goddard Space Flight Center, National Aeronautics and Space Administration (NASA) [35,36]. The time resolution was 1 h; hence, there were a total of 8,784 datasets. Each dataset was the average of the hourly time interval. To evaluate the feasibility and generality of this flexible operation strategy, 10 cities located in China were selected for comparison. Because China accounts for one-third of the world's ammonia production and half of the carbon emissions, energy use in the Chinese ammonia industry is the largest worldwide [37]. The basic information is shown in Table 5.

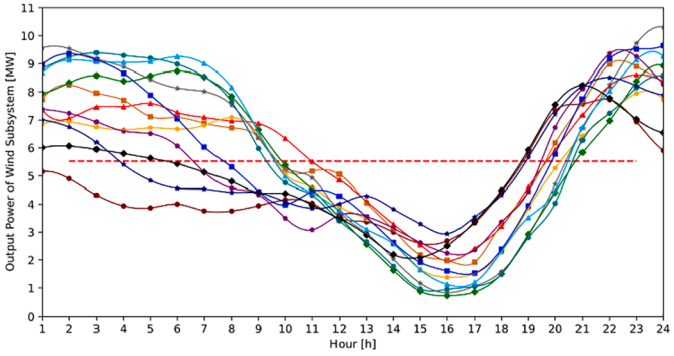
The monthly average of each hour of solar irradiance and total solar energy for each month are shown in Fig. 3 **(a)** and **(b)**, respectively**.** The red dashed line represents the average value for the entire year. The hourly distribution of solar irradiance was bell-shaped, and the peak appeared between 14:00 and 15:00. In the summer from May to July, the peak solar irradiance exceeded 900 W·m^{-2}. In this period, solar energy was abundant, and the daily daylight hours totaled almost 15 h. In the winter from November to January, the solar irradiance decreased to less than 600 W·m^{-2}, solar energy was relatively insufficient, and there were only 11 h of daily daylight. The highest solar energy potential was 25,076 MWh in July, while the lowest was 9,640 MWh in December. The monthly average wind speed and total wind energy for each month are shown in Fig. 3 **(c)** and **(d)**, respectively. The hourly distribution of average daily wind speed was approximately valley-shaped with the valley point appearing at noon when the solar energy increased to a maximum, while the peak appeared in the evening from 19:00 to 23:00 when the solar energy decreased to zero. The wind speed and solar energy complemented each other. In March, May, and August, wind energy was abundant, and the highest wind energy potential was 16,964 MWh in August. Similar to the distribution of total solar energy from December to January, wind energy was lacking, with the lowest



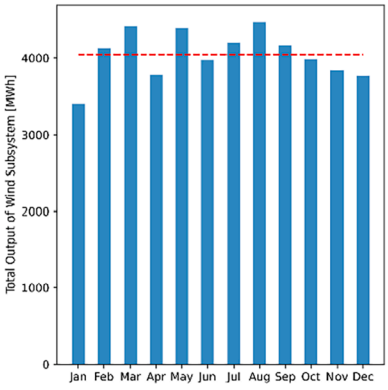
(a) Daily distribution of output power of solar subsystem evaluated on the monthly average of each hour



(b) Total monthly output of solar subsystem



(c) Daily distribution of output power of wind subsystem evaluated on the monthly average of each hour



(d) Total monthly output of wind subsystem

Fig. 4. Energy output of solar and wind subsystems.

potential being 6,246 MWh in December. There was a risk of energy shortage in the winter because both the solar and wind energy were insubstantial.

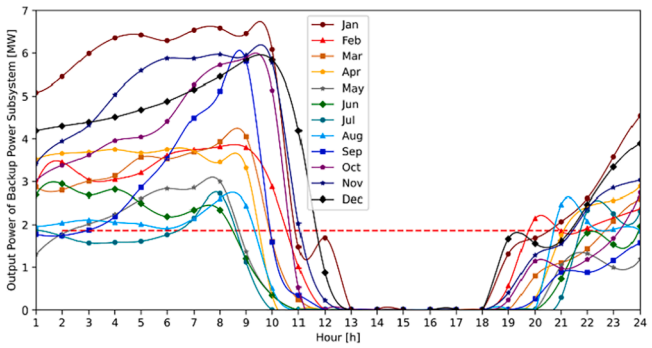
3. Results and discussion

This section presents the performance of the entire system and each subsystem based on thermodynamic analyses and dynamic simulations. The energy input and output of each subsystem, effects of thermal storage and hydrogen storage, ammonia and hydrogen yields, electricity consumption, amount of heat released, energy and exergy efficiencies, and lateral comparison of different cities are shown in detail.

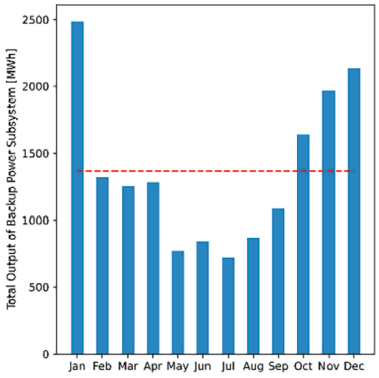
3.1. Energy output of solar and wind subsystems

Because solar energy and wind energy have opposite daily distribution curves, solar and wind subsystems have to provide constant baseload electricity together, including the self-consumed electricity. The monthly average of the hourly output power and the total energy output of each month of the solar subsystem are shown in Fig. 4 **(a)** and **(b)**, respectively. The red dashed line represents the average value for the entire year. The daily distribution curves of output power were similar to those of solar irradiance; Owing to the equipped thermal storage unit, the effective output time was 16 h, which is longer than the number of daylight hours. Even in the evening, the solar subsystem provided a portion of the output power. The maximum output power that could be sustained for 3 h from 15:00 to 18:00 was approximately 12 MW in the summer and decreased to 9 MW in the winter. In addition, the output power exhibited hysteresis characteristics. Starting at 7:00, after saving sufficient hot HTF for power generation, there was a sharp increase in the output power at approximately 9:00, while after 22:00, the thermal storage unit still provided energy as shown by the slow decline in the output power. The solar subsystem continuously provided abundant energy output from May to August. The highest energy output was 3,662 MWh in July. Because of the lack of solar energy in the winter, the energy output was insufficient, and the lowest energy output was 1,905 MWh in December.

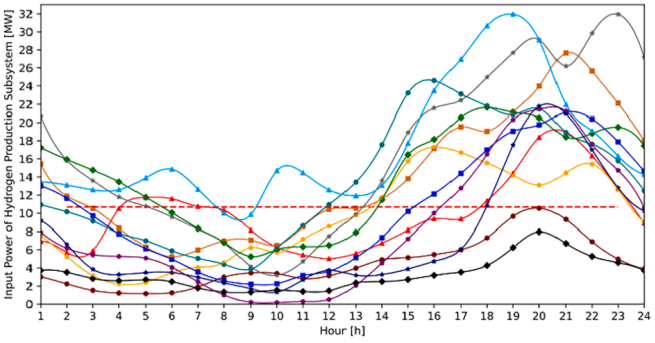
The monthly average output power and total energy output of each month of the wind subsystem are shown in Fig. 4 **(c)** and **(d)**, respectively. As mentioned above, the use of the output power of the solar subsystem was the first priority, and the wind subsystem was used to fulfill the remainder of the baseload. The daily distribution curves of the



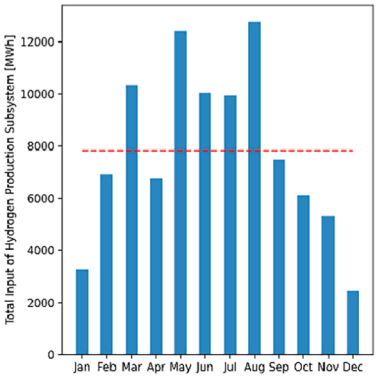
(a) Daily distribution of output power of backup power subsystem evaluated on the monthly average of each hour



(b) Total monthly output of backup power subsystem



(c) Daily distribution of input power of hydrogen production subsystem evaluated on the monthly average of each hour



(d) Total monthly input of hydrogen production subsystem

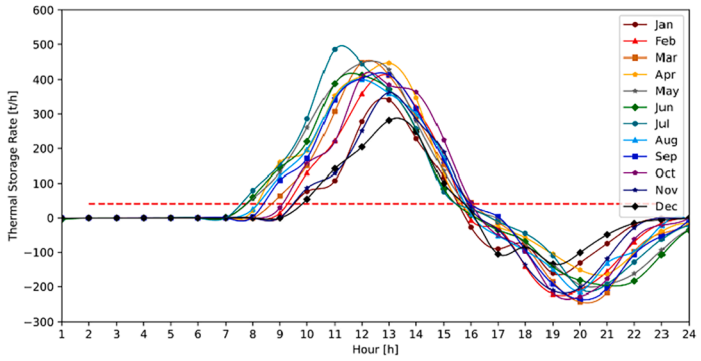
Fig. 5. Energy output and input of backup power and hydrogen production subsystems.

output power exhibited an opposite trend to that of the solar subsystem, particularly in the valley where the solar output determined the wind output. The high output power lasted from 21:00 to 8:00 when there was less or no output power from the solar subsystem. During this period, the wind energy potential determined the output power. The highest output power was approximately 10 MW, while the lowest was below 1 MW. The distribution of the total energy output of the wind subsystem was greater, except for January, and did not coincide with the distribution of the total wind energy because excess energy was used in the hydrogen production subsystem. The highest energy output was 4,474 MWh, and the lowest was 3,410 MWh.

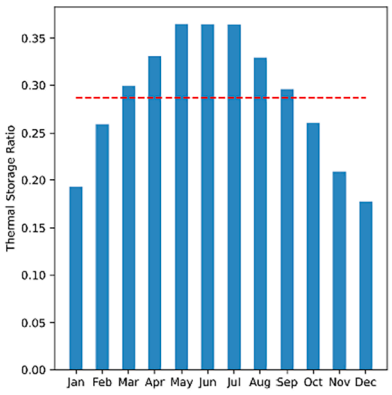
3.2. Energy output of backup power subsystem and excess energy used in hydrogen production subsystem

Hydrogen, as a short-term energy storage medium, is used to store excess energy and release it when there is an energy shortage. Fuel cells are the main part of the backup power subsystem that used hydrogen to generate electricity. This is a supplement to the solar and wind output. The monthly average of the hourly output power and total energy output of each month of the backup power subsystem are shown in Fig. 5 **(a)** and **(b)**, respectively. The red dashed line represents the average value for the entire year. The daily distribution curves of the output power had a peak in the morning between 8:00 and 9:00 when the solar subsystem started to provide output power, and the wind subsystem steeply decreased its output power. The total output power was insufficient during that period. The highest peak output power was approximately 6.5 MW in the winter, and the operating time of the subsystem was 19 h, while the lowest peak output power was approximately 2.5 MW in the summer when the operating time declined to 14 h. After the peak, there was a sharp decrease owing to the rapid increase in the output power of the solar subsystem. In addition, the system was activated after 19:00 when the output power of the solar subsystem sharply decreased. The total energy output of each month of the backup power subsystem was over 2,000 MWh from November to January and demonstrates the energy deficiency in winter. The highest energy output was 2,488 MWh in January, and the lowest was 724 MWh in July. To provide a constant electricity baseload, the backup power subsystem provided more output power in the winter, but the hydrogen stored and produced by the system itself could not fulfill this requirement. Extra hydrogen needed to be imported into the system, or, if there was a long-term energy storage medium such as ammonia, it could be used during this period to offset the deficiency of hydrogen.

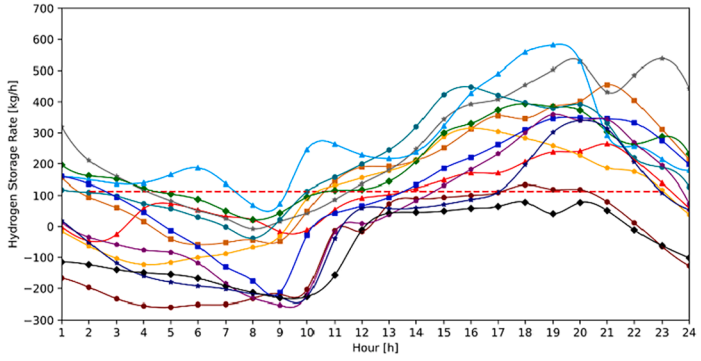
Water electrolysis is the main part of the hydrogen production subsystem, where excess energy is utilized to produce hydrogen. The monthly average hourly input power and total monthly energy input of



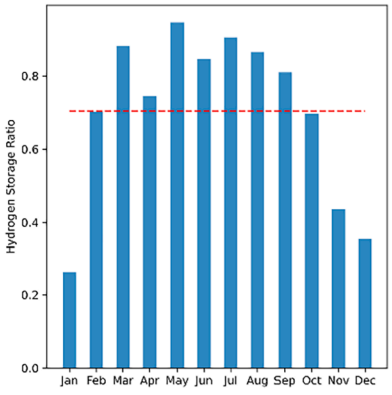
(a) Daily distribution of thermal storage rate evaluated on the monthly average of each hour



(b) Average monthly thermal storage ratios



(c) Daily distribution of hydrogen storage rate evaluated on the monthly average of each hour



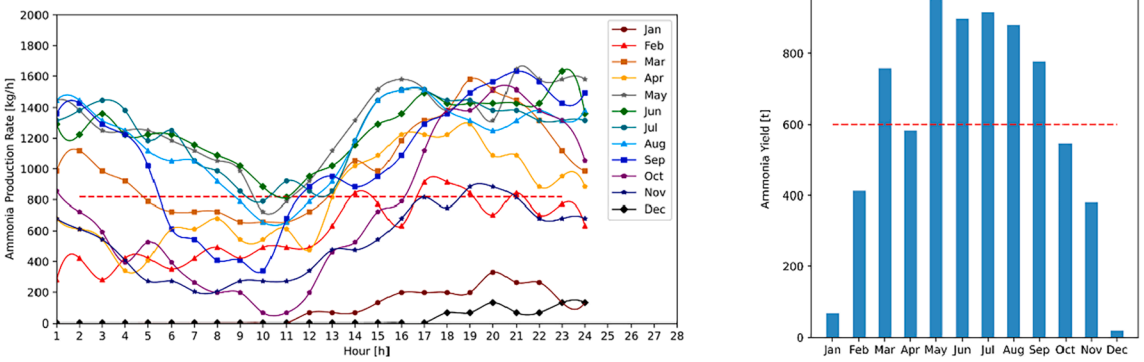
(d) Average monthly hydrogen storage ratios

Fig. 6. Production rate and storage ratio of thermal and hydrogen storage units.

the hydrogen production subsystem are shown in Fig. 5 **(c)** and **(d)**, respectively. The results were the same as those of the distribution of excess energy. The daily distribution curves of the input power reached a maximum in the evening when wind energy was abundant. The advanced peak in July was owing to the prominent influence of excess solar energy. Compared to Fig. 5 **(a)**, between 8:00 and 12:00, when the output power of the backup power subsystem reached a maximum and then decreased to zero, the input power of the hydrogen production subsystem reached a minimum. The lowest input power decreased to zero in October. From 13:00 to 18:00, when the output power of the backup power subsystem was near zero, the excess energy started to increase. The sufficient output power of the solar subsystem ensured that there was an increase in excess energy, while the output power of the wind subsystem was the main part. In May and August, when the wind energy was adequate, the highest input power was approximately 32 MW, but the input power was below the average in December and January. The lowest peak value declined to 8 MW, which is a quarter of the highest value. The distribution of the total energy input was similar to that of wind energy, which proves that wind energy provided the majority of the excess energy. The highest energy input was 12,750 MWh in August, and the lowest was 2,475 MWh in December. The huge difference within one day and between different months necessitates the storage of energy for later use. However, a short-term energy storage medium cannot cover this huge gap between different months; therefore, it is imperative to introduce a long-term energy storage medium such as ammonia into the system.

3.3. *Thermal storage and hydrogen storage*

The thermal storage unit plays an important role in providing constant baseload electricity. By controlling the mass flow rates in the receiver and heat exchanger loops, the solar subsystem can provide stable and durable power output regardless the fluctuation of solar irradiance. The monthly average hourly thermal storage rate and average monthly thermal storage ratios are shown in Fig. 6 **(a)** and **(b)**, respectively. The red dashed line represents the average value for the entire year. The daily distribution curves were approximately sine shaped. Starting from 7:00, the thermal storage unit began to store heat, and the thermal storage rate increased to a maximum between 11:00 and 13:00. In this period, because the output power of the solar subsystem did not reach the maximum and the solar energy continuously increased, both the storage rate and its gradient were positive. The



(a) Daily distribution of ammonia production rate evaluated on the monthly average of each hour

(b) Monthly ammonia yield

Fig. 7. Production rate and total yield of ammonia synthesis subsystem.

highest thermal storage rate was approximately 500 t·h^{-1} in July. Between 14:00 and 16:00, the output power of the solar subsystem sustained a high level and the storage rate was positive, but its gradient became negative. The heat-accumulation process ensured that the curves of the thermal storage rate were above zero. To maintain a high level of output power, the thermal storage rate constantly decreased until 19:00. The negative storage rate and its gradient were the result of a decrease in solar energy. After 19:00, the output power in the solar subsystem decreased sharply, and the thermal storage rate gradually became zero at 24:00, which means that the solar subsystem stopped functioning. The curves of the thermal storage rate were below zero because of the heat-releasing process. The average thermal storage ratio for each month represents the excess solar energy potential that can be used. This distribution was similar to that of the solar energy, which from May to July had a relatively high potential, and the average thermal storage ratio was over 35%, while from November to January, it had a relatively low potential and the ratio was lower than 20%.

Hydrogen was used as a short-term energy storage medium to solve the uneven distribution of renewable energy within one day. The monthly average hourly hydrogen storage rate and monthly average hydrogen storage ratios are shown in Fig. 6 **(c)** and **(d)**, respectively. The daily distribution curves of the hydrogen storage rate reached a peak when the input power of the hydrogen production subsystem was at its maximum, while it decreased to the valley point when the output power of the backup power subsystem was at its maximum. Both subsystems influenced the distribution curves, particularly when they achieved maximum power. Before 10:00, the backup power subsystem consumed more hydrogen.Most of the curves were below the average line, after which the hydrogen production subsystem started to produce more hydrogen and the hydrogen storage rate became positive. The maximum storage rate was over 500 kg·h^{-1} in May and August when wind energy was abundant. The minimum was below −250 kg·h^{-1} in January when relatively less renewable energy was available. Except for November, December, and January, the average hydrogen storage ratio was over 70%, which means that at least 17 h of constant baseload electricity was provided. The storage ratio below 40% in December and January was attributed to the low renewable energy potential; less hydrogen was produced and more was consumed. Thus, it is best to perform equipment maintenance during this period to mitigate the hydrogen shortage.

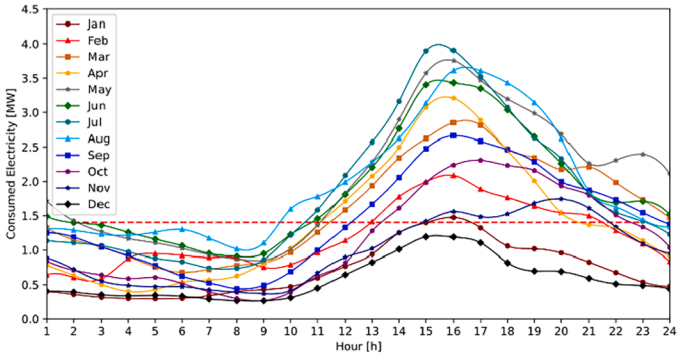
3.4. Ammonia production

Hydrogen cannot be stored without limits for safety and cost. Thus, it is difficult to use the excess energy of summer during winter without other options. As a long-term energy storage medium, ammonia can be used to solve the uneven distribution of renewable energy between different months. When the amount of stored hydrogen exceeds the limit of satisfying the 24-h demand, the excess hydrogen will be sent to the ammonia synthesis subsystem. Ammonia is synthesized periodically at a constant rate when there is a sufficient amount of hydrogen for the constant production of ammonia. The monthly average hourly ammonia production rate and total monthly yields are shown in Fig. 7 **(a)** and **(b)**, respectively. The red dashed line represents the average value for the entire year. The daily distribution curves were V-shaped, and most of the curves reached the valley point between 10:00 and 11:00 and then increased to the maximum in the evening. This trend was similar to that of the hydrogen storage rate but fluctuated more because ammonia was produced regularly. The highest production rate was approximately 1,700 kg·h^{-1} from May to August. Except for January and December, the total monthly yield was considerable. The largest ammonia yield was 960 t in May. This distribution was similar to that of wind energy. Ammonia synthesis can stop in December and January to save hydrogen because a larger amount of hydrogen is required in the backup power subsystem and renewable energy is relatively insufficient in this period. In addition, ammonia produced in the summer can be used as a fuel to offset the hydrogen shortage in winter.

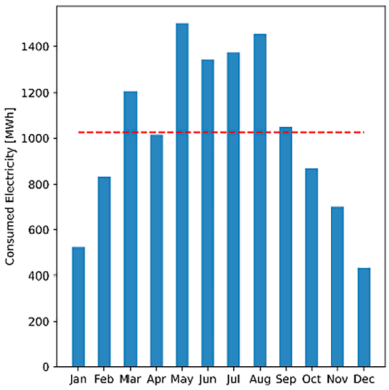
3.5. Consumed electricity and released heat

The monthly averages of hourly and total consumed electricity are shown in Fig. 8 **(a)** and **(b)**, respectively. The red dashed line represents the average value for the entire year. Consumed electricity comes from pumps in the solar subsystem, compressors, and the PSA unit in the hydrogen production and ammonia synthesis subsystems. The daily distribution curves were inverted and V-shaped. The peak distribution appeared between 15:00 and 16:00 when the output power of the solar subsystem was at its maximum, and the input power of the hydrogen production subsystem remained at a high level. The maximum was approximately 4 MW in July. After 22:00, the output power of the solar subsystem gradually decreased to zero, and the compressors and PSA unit consumed the majority ofthe main part of the consumed electricity. The highest total monthly electricity consumed was 1,502 MWh in May, and the lowest was 432 MWh in December.

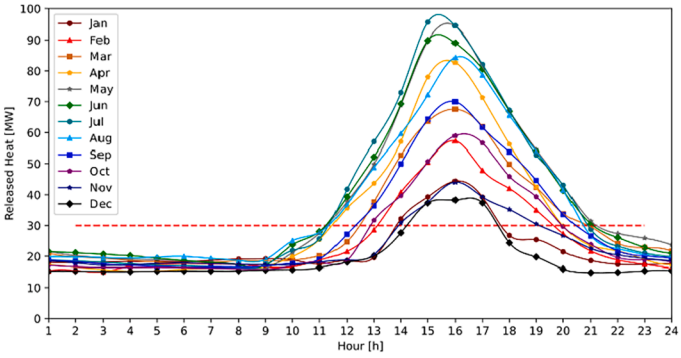
The monthly hourly average and total monthly released heat are shown in Fig. 8 **(c)** and **(d)**, respectively. The released heat came from the Rankine cycle, ammonia synthesis reactor, compressors, water



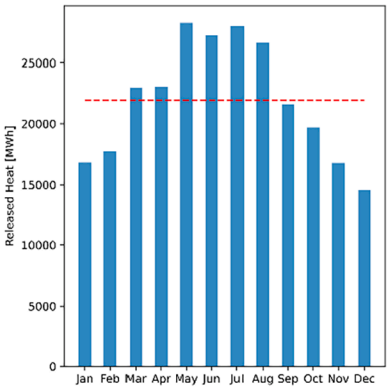
(a) Daily distribution of consumed electricity evaluated on the monthly average of each hour



(b) Total electricity consumed monthly



(c) Daily distribution of released heat evaluated on the monthly average of each hour



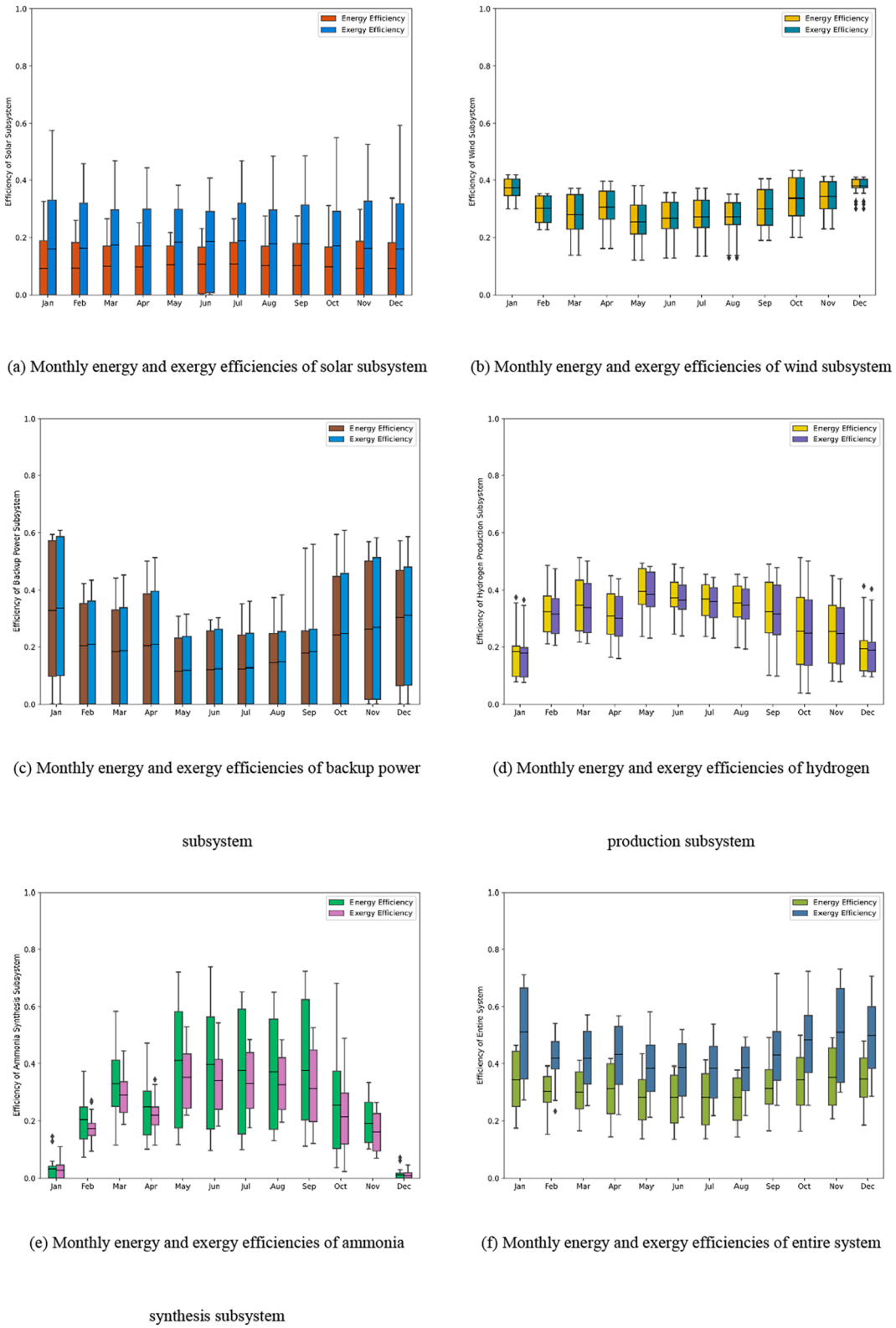
(d) Total amount of heat released monthly

Fig. 8. Consumed electricity and released heat.

electrolysis, and fuel cells. The inverted V-shaped daily distribution curves were mostly attributed to the waste heat from the Rankine cycle. The released heat reached a peak between 15:00 and 16:00 when the output power of the solar subsystem was at a maximum. From April to August, the released heat remained at a high level. The highest peak value was approximately 100 MW in July. In contrast, in winter, owing to the decrease in the average output power of the solar subsystem, it decreased to approximately 40 MW. Other components provided an approximately constant heat load as shown in the flattened curve between 1:00 and 9:00. The distribution of the total monthly released heat was similar to that of the solar energy. The largest monthly released heat was 28,300 MWh in May, and the lowest was 14,572 MWh in December. The daily and monthly distribution of released heat was not good for district heating. The heat demand was high during the evening and winter. Therefore, it is better to adjust the operation strategy by considering the heat demand. In summer, the heat demand is low, and the released heat can be used for power generation and the further storage of excess energy in ammonia. In winter, when the heat demand is high, ammonia can be burned for heat directly. Moreover, ammonia synthesis can be centralized in the evening to alleviate the uneven distribution of the heat load within one day. The only cost of this change is a larger hydrogen storage tank. Ammonia, as a long-term energy storage medium, can solve not only the seasonal uneven distribution of renewable energy but also the heat demand.

3.6. Energy and exergy efficiencies of entire system and its subsystems

Fig. 9 **(a)–(e)** show the box plots of the energy and exergy efficiencies of each subsystem. Each box shows the maximum, minimum, and average efficiencies that vary depending on the output. The range of each box represents the degree of fluctuation of the output. Energy storage units have a significant effect on the change in efficiency in a single time interval. In the solar subsystem, the energy efficiency changed from 0% to 33.7%, while the exergy efficiency varied from 0% to 59.4%. The average efficiencies for each month were similar. The distribution of the maximum efficiencies was opposite to that of the total output of the solar subsystem. The high energy and exergy efficiencies appeared in December and January because excess energy was not considered in Eqs. (32 and 33). When the output power exceeded the demand, more power was provided and the efficiency was lower. In addition, the use of waste heat was not considered, which reduces the efficiencies. In the wind subsystem, the energy and exergy efficiencies



(a) Monthly energy and exergy efficiencies of solar subsystem

(b) Monthly energy and exergy efficiencies of wind subsystem

(c) Monthly energy and exergy efficiencies of backup power subsystem

(d) Monthly energy and exergy efficiencies of hydrogen production subsystem

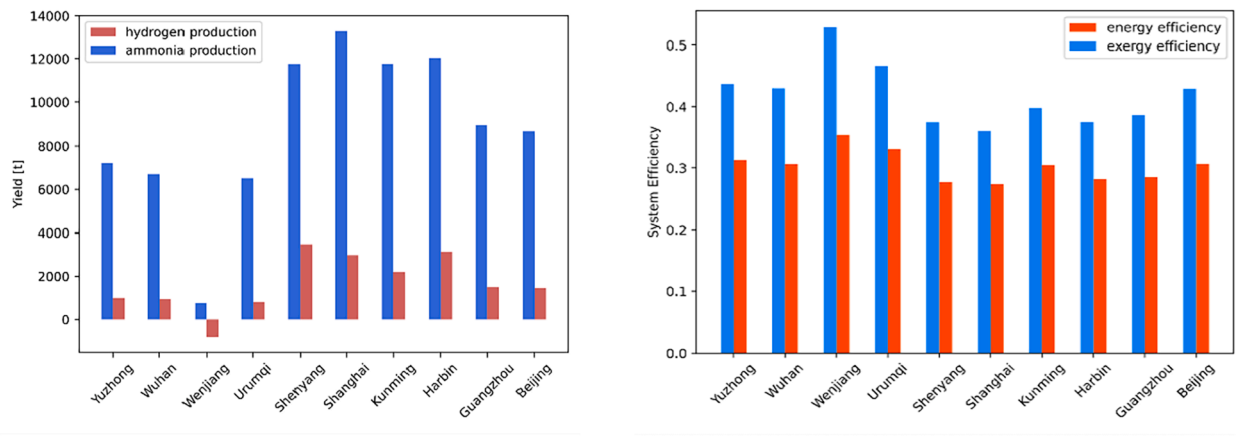
(e) Monthly energy and exergy efficiencies of ammonia synthesis subsystem

(f) Monthly energy and exergy efficiencies of entire system

Fig. 9. Energy and exergy efficiencies of entire system and each subsystem.

changed from 12% to 43.5%, and maximum efficiencies were observed in October. The distribution of the average efficiencies was opposite of that of the total output of the solar subsystem because the wind subsystem offset the deficiency of the output power of the solar subsystem. A larger deficiency results in higher efficiencies of the wind subsystem. Moreover, the excess energy of the wind subsystem used in water electrolysis was not considered. The distribution of minimum efficiencies had a negative correlation with the input power of the hydrogen production subsystem.

In the backup power subsystem, the energy efficiency changed from 0% to 59.5%, while the exergy efficiency varied from 0% to 60.9%. The efficiency is related to the output power. The subsystem provided high



(a) Yearly hydrogen and ammonia yields of systems in each city (b) Average energy and exergy efficiencies of systems in each city

Fig. 10. Comparison of different cities based on system performance.

output power from October to January; both the maximum and average efficiencies were at a high level during that period. In the hydrogen production subsystem, the energy efficiency changed from 3.9% to 51.3%, while the exergy efficiency variance from 3.9% to 50.1% was the opposite distribution as that of the backup power subsystem. In the ammonia synthesis subsystem, the energy efficiency changed from 0% to 73.8%, while the exergy efficiency varied from 0% to 54.3%. From May to September, the efficiencies remained high because of sufficient energy and hydrogen. In other months, production should be centralized to improve efficiency. Fig. 9 **(f)** shows the total efficiency. The total energy efficiency ranged from 13.5% to 50%, while the total exergy efficiency ranged from 21.2% to 73.1%. This significant difference was attributed to the total output power of the wind subsystem. It is obvious that in the month with more solar energy production, the total efficiencies are relatively low. Therefore, increasing the efficiencies of the solar subsystem, such as the efficient use of waste heat, is an effective way to improve the total energy and exergy efficiencies. In addition, optimizing the timing of ammonia production and regulating the heat supply can improve the total efficiency.

3.7. Comparison of system performance in different cities

Ten cities in China were selected to evaluate the system performance in different regions. Fig. 10 **(a)** shows the hydrogen and ammonia yields. Hydrogen and ammonia are the two energy storage media that increase the flexibility of system operation. Hydrogen relieves the uneven daily distribution of renewable energy by storing excess energy and releasing it during an energy shortage. The use of ammonia alleviates the non-uniform monthly distribution of renewable energy and provides a safe hydrogen storage method. In addition, both can be transported to other areas to provide benefits. The highest hydrogen yield was 3,746 t in Shenyang, and the highest ammonia yield was 13,300 t in Shanghai. Other cities, such as Kunming and Harbin, also had considerable hydrogen and ammonia yields. High hydrogen and ammonia yields indicate that they have the potential to generate more electricity and generate more profit. These four cities are more suitable for this renewable multi-generation system and flexible storage strategy. Fig. 10 **(b)** shows the energy and exergy efficiencies of the system in different cities. As mentioned previously, the difference in efficiencies is caused by the ratio of the output power of the wind subsystem. Relatively low efficiencies are caused by the solar subsystem utilizing a significant amount of the total output power. Although the systems in Wenjiang and Urumqi had higher efficiencies, the lower hydrogen and ammonia yields reduced their cost effectiveness.

4. Conclusions

In this study, an integrated renewable multi-generation system and flexible operation strategy were designed to provide a constant baseload electricity of 10 MW. Hydrogen and ammonia as short- and long-term energy storage media were used to solve the uneven daily and monthly distribution of renewable energy. A thermodynamic model of the entire system was constructed to evaluate the system performance. In addition, dynamic simulations and analyses were conducted based on the hourly weather data of 10 cities in China in 2020 to examine the feasibility and generality of the system. Yuzhong was selected for the demonstration of system performance, including the input and output of each subsystem, the effects of energy storage units, electricity consumption, amount of released heat, and efficiencies. The main conclusions are as follows.

1. The maximum energy and exergy efficiencies of the entire system were 50% and 73.1%, respectively.
2. The total energy output of the solar subsystem was 33,285.6 MWh, and the maximum energy and exergy efficiencies were 33.7% and 59.4%, respectively. The total energy output of the wind subsystem was 48,535.9 MWh, and the maximum energy and exergy efficiencies were 43.5%. The total energy output of the backup power subsystem was 16,384.9 MWh, and the maximum energy and exergy efficiencies were 59.5% and 60.9%, respectively.
3. The net hydrogen yield of the hydrogen production subsystem was 994.5 $t \cdot y^{-1}$, and the maximum energy and exergy efficiencies were 51.3% and 50.1%, respectively. The ammonia yield of the ammonia synthesis subsystem was 7201 $t \cdot y^{-1}$, and the maximum energy and exergy efficiencies were 73.8% and 54.3%, respectively.
4. This system has great potential in Shenyang, Shanghai, Kunming, and Harbin with its larger hydrogen and ammonia yields.
5. It is recommended to use waste heat as much as possible to improve the solar efficiency and total efficiency. The operation strategy should consider the mismatch of heat supply and demand and hydrogen shortage in winter by means of waste heat utilization and ammonia synthesis. It is proposed to perform equipment maintenance in January because of the lower renewable energy potential and system efficiency at that time.

CRediT authorship contribution statement

Du Wen: Conceptualization, Methodology, Software, Writing – original draft. **Muhammad Aziz:** Supervision, Validation, Resources,

Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The thermodynamic analysis of each subsystem including the mass, enthalpy, entropy, and exergy balance can be summarized as follows:
Mass balance in solar subsystem:

$$\int \dot{m}_1 dt = \int \dot{m}_2 dt = \int \dot{m}_3 dt = \int \dot{m}_7 dt \tag{40}$$

$$\int \dot{m}_4 dt = \int \dot{m}_5 dt = \int \dot{m}_6 dt \tag{41}$$

$$\int \dot{m}_8 dt = \int \dot{m}_9 dt = \int \dot{m}_{10} dt = \int \dot{m}_{11} dt \tag{42}$$

Parabolic trough collector:

$$\int \dot{m}_2 h_2 dt + \int \dot{Q}_{\text{solar}} dt = \int \dot{m}_3 h_3 dt \tag{43}$$

$$\int \dot{m}_2 s_2 dt + \int \frac{\dot{Q}_{\text{solar}}}{T_{\text{r}}} dt + \int \dot{S}_{\text{gen, r}} dt = \int \dot{m}_3 s_3 dt \tag{44}$$

$$\int \dot{m}_2 ex_2 dt + \int \dot{Q}_{\text{solar}} \left(1 - \frac{T_{\text{a}}}{T_{\text{r}}}\right) dt = \int \dot{m}_3 ex_3 dt + \int \dot{E}x_{\text{dest, r}} dt \tag{45}$$

Thermal storage tank:

$$\int \dot{m}_3 h_3 dt + \int \dot{m}_6 h_6 dt = \int \dot{m}_4 h_4 dt + \int \dot{m}_7 h_7 dt \tag{46}$$

$$\int \dot{m}_3 s_3 dt + \int \dot{m}_6 s_6 dt + \int \dot{S}_{\text{gen, ts}} dt = \int \dot{m}_4 s_4 dt + \int \dot{m}_7 s_7 dt \tag{47}$$

$$\int \dot{m}_3 ex_3 dt + \int \dot{m}_6 ex_6 dt = \int \dot{m}_4 ex_4 dt + \int \dot{m}_7 ex_7 dt + \int \dot{E}x_{\text{dest, ts}} dt \tag{48}$$

Heat Exchanger 1:

$$\int \dot{m}_4 h_4 dt + \int \dot{m}_8 h_8 dt = \int \dot{m}_5 h_5 dt + \int \dot{m}_9 h_9 dt \tag{49}$$

$$\int \dot{m}_4 s_4 dt + \int \dot{m}_8 s_8 dt + \int \dot{S}_{\text{gen, hx1}} dt = \int \dot{m}_5 s_5 dt + \int \dot{m}_9 s_9 dt \tag{50}$$

$$\int \dot{m}_4 ex_4 dt + \int \dot{m}_8 ex_8 dt = \int \dot{m}_5 ex_5 dt + \int \dot{m}_9 ex_9 dt + \int \dot{E}x_{\text{dest, hx1}} dt \tag{51}$$

Pump 1:

$$\int \dot{m}_1 h_1 dt + \int \dot{W}_{\text{p1}} dt = \int \dot{m}_2 h_2 dt \tag{52}$$

$$\int \dot{m}_1 s_1 dt + \int \dot{S}_{\text{gen, p1}} dt = \int \dot{m}_2 s_2 dt \tag{53}$$

$$\int \dot{m}_1 ex_1 dt + \int \dot{W}_{\text{p1}} dt = \int \dot{m}_2 ex_2 dt + \int \dot{E}x_{\text{dest, p1}} dt \tag{54}$$

Pump 2:

$$\int \dot{m}_5 h_5 dt + \int \dot{W}_{\text{p2}} dt = \int \dot{m}_6 h_6 dt \tag{55}$$

$$\int \dot{m}_5 s_5 dt + \int \dot{S}_{\text{gen, p2}} dt = \int \dot{m}_6 s_6 dt \tag{56}$$

$$\int \dot{m}_5 ex_5 dt + \int \dot{W}_{\text{p2}} dt = \int \dot{m}_6 ex_6 dt + \int \dot{E}x_{\text{dest, p2}} dt \tag{57}$$

Rankine cycle:

$$\int \dot{m}_9 h_9 dt = \int \dot{m}_{10} h_{10} dt + \int \dot{W}_t dt \tag{58}$$

$$\int \dot{m}_9 s_9 dt + \int \dot{S}_{\text{gen, t}} dt = \int \dot{m}_{10} s_{10} dt \tag{59}$$

$$\int \dot{m}_9 ex_9 dt = \int \dot{m}_{10} ex_{10} dt + \int \dot{W}_t dt + \int \dot{E}x_{\text{dest, t}} dt \tag{60}$$

Heat Exchanger 2:

$$\int \dot{m}_{10} h_{10} dt + \int \dot{m}_{25} h_{25} dt = \int \dot{m}_{11} h_{11} dt + \int \dot{m}_{26} h_{26} dt + \int \dot{Q}_{\text{hx2}} dt \tag{61}$$

$$\int \dot{m}_{10} s_{10} dt + \int \dot{m}_{25} s_{25} dt + \int \dot{S}_{\text{gen, hx2}} dt = \int \dot{m}_{11} s_{11} dt + \int \dot{m}_{26} s_{26} dt + \int \frac{\dot{Q}_{\text{hx2}}}{T_{\text{hx2}}} dt \tag{62}$$

$$\int \dot{m}_{10} ex_{10} dt + \int \dot{m}_{25} ex_{25} dt = \int \dot{m}_{11} ex_{11} dt + \int \dot{m}_{26} ex_{26} dt + \int \dot{Q}_{\text{hx2}} (1 - \frac{T_{\text{a}}}{T_{\text{hx2}}}) dt + \int \dot{E}x_{\text{dest, hx2}} dt \tag{63}$$

Pump 3:

$$\int \dot{m}_{11} h_{11} dt + \int \dot{W}_{\text{p3}} dt = \int \dot{m}_8 h_8 dt \tag{64}$$

$$\int \dot{m}_{11} s_{11} dt + \int \dot{S}_{\text{gen, p3}} dt = \int \dot{m}_8 s_8 dt \tag{65}$$

$$\int \dot{m}_{11} ex_{11} dt + \int \dot{W}_{\text{p3}} dt = \int \dot{m}_8 ex_8 dt + \int \dot{E}x_{\text{dest, p3}} dt \tag{66}$$

Mass balance in wind subsystem:

$$\int \dot{m}_{13} dt = \int \dot{m}_{18} dt \tag{67}$$

$$\int \dot{m}_{14} dt + \int \dot{m}_{16} dt = \int \dot{m}_{19} dt \tag{68}$$

$$\int \dot{m}_{17} dt = \int \dot{m}_{20} dt \tag{69}$$

Air separator:

$$\int \dot{m}_{\text{air}} h_{\text{air}} dt + \int \dot{W}_{\text{PSA}} dt = \int \dot{m}_{13} h_{13} dt + \int \dot{m}_{14} h_{14} dt + \int \dot{Q}_{\text{PSA}} dt \tag{70}$$

$$\int \dot{m}_{\text{air}} s_{\text{air}} dt + \int \dot{S}_{\text{gen, PSA}} dt = \int \dot{m}_{13} s_{13} dt + \int \dot{m}_{14} s_{14} dt + \int \frac{\dot{Q}_{\text{PSA}}}{T_{\text{PSA}}} dt \tag{71}$$

$$\int \dot{m}_{\text{air}} ex_{\text{air}} dt + \int \dot{W}_{\text{PSA}} dt = \int \dot{m}_{13} ex_{13} dt + \int \dot{m}_{14} ex_{14} dt + \int \dot{Q}_{\text{PSA}} \left(1 - \frac{T_a}{T_{\text{PSA}}}\right) dt + \int \dot{E}x_{\text{dest, PSA}} dt \tag{72}$$

Compressor 1:

$$\int \dot{m}_{13} h_{13} dt + \int \dot{W}_{\text{c1}} dt = \int \dot{m}_{18} h_{18} dt + \int \dot{Q}_{\text{c1}} dt \tag{73}$$

$$\int \dot{m}_{13} s_{13} dt + \int \dot{S}_{\text{gen, c1}} dt = \int \dot{m}_{18} s_{18} dt + \int \frac{\dot{Q}_{\text{c1}}}{T_{\text{c1}}} dt \tag{74}$$

$$\int \dot{m}_{13} ex_{13} dt + \int \dot{W}_{\text{c1}} dt = \int \dot{m}_{18} ex_{18} dt + \int \dot{Q}_{\text{c1}} \left(1 - \frac{T_a}{T_{\text{c1}}}\right) dt + \int \dot{E}x_{\text{dest, c1}} dt \tag{75}$$

Compressor 2:

$$\int \dot{m}_{14} h_{14} dt + \int \dot{m}_{16} h_{16} dt + \int \dot{W}_{\text{c2}} dt = \int \dot{m}_{19} h_{19} dt + \int \dot{Q}_{\text{c2}} dt \tag{76}$$

$$\int \dot{m}_{14} s_{14} dt + \int \dot{m}_{16} s_{16} dt + \int \dot{S}_{\text{gen, c2}} dt = \int \dot{m}_{19} s_{19} dt + \int \frac{\dot{Q}_{\text{c2}}}{T_{\text{c2}}} dt \tag{77}$$

$$\int \dot{m}_{14} ex_{14} dt + \int \dot{m}_{16} ex_{16} dt + \int \dot{W}_{\text{c2}} dt = \int \dot{m}_{19} ex_{19} dt + \int \dot{Q}_{\text{c2}} \left(1 - \frac{T_a}{T_{\text{c2}}}\right) dt + \int \dot{E}x_{\text{dest, c2}} dt \tag{78}$$

Table 6
Thermophysical states at each point in the system on the designed condition.

—	T[i] [K]	P[i] [kPa]	h[i] [kJ·kg^{-1}]	s[i] [kJ·kg^{-1}·K^{-1}]	ex[i] [kJ·kg^{-1}]	—	T[i] [K]	P[i] [kPa]	h[i] [kJ·kg^{-1}]	s[i] [kJ·kg^{-1}·K^{-1}]	ex[i] [kJ·kg^{-1}]
1	561.2	1,195	−3,037	3.666	134.1	15	353.2	101.3	335.1	1.076	18.94
2	563.2	1,200	−3,035	3.669	135.1	16	293.2	101.3	−4.593	−0.01548	0.03972
3	838.2	1,195	−2,754	4.074	295.1	17	293.2	101.3	3,860	53.13	0.6061
4	838.2	1,195	−2,754	4.074	295.1	18	293.2	15,000	276	5.241	442.1
5	559.2	1,190	−3,039	3.662	133	19	293.2	15,000	−40.1	−1.413	381.2
6	561.2	1,195	−3,037	3.666	134.1	20	293.2	15,000	3,932	32.39	6256
7	561.2	1,195	−3,037	3.666	134.1	21	293.2	7,500	288.3	5.49	380.3
8	300.2	13,000	125.1	0.3915	12.93	22	353.2	100	50.8	0.1598	3.176
9	823.2	13,000	3,471	6.609	1,506	23	353.2	100	4,723	55.86	49.26
10	485.2	1,046	2,855	6.731	8,52.3	24	293.2	7,500	3,893	35.31	5,348
11	298.2	1,046	105.8	0.367	0.9471	25	293.2	15,000	921.1	10.03	1,468
12	353.2	101.3	335.1	1.076	18.94	26	473.2	15,000	1,560	11.73	1,600
13	293.2	31.6	304.2	7.164	−103	27	293.2	12,000	299.5	1.286	337.6
14	293.2	8.4	−4.357	0.6321	−192.8						

Compressor 3:

$$\int \dot{m}_{17}h_{17}dt + \int \dot{W}_{c3}dt = \int \dot{m}_{20}h_{20}dt + \int \dot{Q}_{c3}dt \tag{79}$$

$$\int \dot{m}_{17}s_{17}dt + \int \dot{S}_{\text{gen, c3}}dt = \int \dot{m}_{20}s_{20}dt + \int \frac{\dot{Q}_{c3}}{T_{c3}}dt \tag{80}$$

$$\int \dot{m}_{17}ex_{17}dt + \int \dot{W}_{c3}dt = \int \dot{m}_{20}ex_{20}dt + \int \dot{Q}_{c3}\left(1-\frac{T_a}{T_{c3}}\right)dt + \int \dot{E}x_{\text{dest, c3}}dt \tag{81}$$

Mass balance in hydrogen production subsystem:

$$\int \dot{m}_{15}dt = \int \dot{m}_{16}dt + \int \dot{m}_{17}dt \tag{82}$$

PEM water electrolysis:

$$\int \dot{m}_{15}h_{15}dt + \int \dot{W}_{\text{PEM}}dt = \int \dot{m}_{16}h_{16}dt + \int \dot{m}_{17}h_{17}dt + \int \dot{Q}_{\text{PEM}}dt \tag{83}$$

$$\int \dot{m}_{15}s_{15}dt + \int \dot{S}_{\text{gen, PEM}}dt = \int \dot{m}_{16}s_{16}dt + \int \dot{m}_{17}s_{17}dt + \int \frac{\dot{Q}_{\text{PEM}}}{T_{15}}dt \tag{84}$$

$$\int \dot{m}_{15}ex_{15}dt + \int \dot{W}_{\text{PEM}}dt = \int \dot{m}_{16}ex_{16}dt + \int \dot{m}_{17}ex_{17}dt + \int \dot{Q}_{\text{PEM}}\left(1-\frac{T_a}{T_{15}}\right)dt + \int \dot{E}x_{\text{dest, PEM}}dt \tag{85}$$

Mass balance in backup power subsystem:

$$\int \dot{m}_{22}dt + \int \dot{m}_{23}dt = \int \dot{m}_{12}dt \tag{86}$$

Fuel cells:

$$\int \dot{m}_{22}h_{22}dt + \int \dot{m}_{23}h_{23}dt = \int \dot{m}_{12}h_{12}dt + \int \dot{W}_{\text{cell}}dt + \int \dot{Q}_{\text{cell}}dt \tag{87}$$

$$\int \dot{m}_{22}s_{22}dt + \int \dot{m}_{23}s_{23}dt + \int \dot{S}_{\text{gen, cell}}dt = \int \dot{m}_{12}s_{12}dt + \int \frac{\dot{Q}_{\text{cell}}}{T_{\text{cell}}}dt \tag{88}$$

$$\int \dot{m}_{22}ex_{22}dt + \int \dot{m}_{23}ex_{23}dt = \int \dot{m}_{12}ex_{12}dt + \int \dot{W}_{\text{cell}}dt + \int \dot{Q}_{\text{cell}}\left(1-\frac{T_a}{T_{\text{cell}}}\right)dt + \int \dot{E}x_{\text{dest, cell}}dt \tag{89}$$

Mass balance in ammonia synthesis subsystem:

$$\int \dot{m}_{21}dt + \int \dot{m}_{24}dt = \int \dot{m}_{25}dt = \int \dot{m}_{26}dt = \int \dot{m}_{27}dt \tag{90}$$

Ammonia synthesis reactor:

$$\int \dot{m}_{26}h_{26}dt = \int \dot{m}_{27}h_{27}dt + \int \dot{Q}_{\text{ASR}}dt \tag{91}$$

$$\int \dot{m}_{26}s_{26}dt + \int \dot{S}_{\text{gen, ASR}}dt = \int \dot{m}_{27}s_{27}dt + \int \frac{\dot{Q}_{\text{ASR}}}{T_{\text{ASR}}}dt \tag{92}$$

Table 7
Working states of system components.

Component	Power [kW]	Heat duty [kW]	Entropy generation [kJ·kg^{-1}·K^{-1}]	Exergy destruction [kW]
Parabolic trough collector	–	−80,970 (in)	1.24	369.6
Steam turbine	15,130	–	2.99	892.4
PEM water electrolysis	37,700	9,542	41.91	12,499
Fuel cell	(10,000)	(8,874)	(14.69)	(4,377)
Ammonia synthesis reactor	–	1,865	2.14	634.1

$$\int \dot{m}_{26} ex_{26} dt = \int \dot{m}_{27} ex_{27} dt + \int \dot{Q}_{\mathrm{ASR}} \left(1 - \frac{T_{\mathrm{a}}}{T_{\mathrm{ASR}}}\right) dt + \int \dot{E}x_{\mathrm{dest,\ ASR}} dt \tag{93}$$

The thermophysical states of each point in the system are listed in Table 6 for the designed condition, while the evaluation of the main components is shown in Table 7. The working states of the main components were achieved under the maximum output condition, and the maximum outputs of the fuel cells and other subsystem did not occur simultaneously.

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